

2024,

Q1. (a) Common drain amplifier

$$(b) V_G = \frac{R_2}{R_1 + R_2} V_{DD} = \frac{6}{4+6} \times 3 = 1.8 \text{ V}$$

$$I_{DS} = \frac{\beta}{2} (V_{GS} - V_{th})^2$$

$$= \frac{\beta}{2} (V_G - V_S - V_{th})^2$$

$$= \frac{\beta}{2} (V_G - I_{DS} R_S - V_{th})^2$$

$$\Rightarrow \pm \sqrt{\frac{2I_{DS}}{\beta}} = V_G - I_{DS} R_S - V_{th}$$

$$\Rightarrow R_S = \frac{V_G - V_{th} \pm \sqrt{\frac{2I_{DS}}{\beta}}}{I_{DS}} = 1k \text{ or } \underline{12k} \Omega$$

$$\text{If } R_S = 1k, V_{GS} = V_G - I_{DS} R_S = 0.4 \text{ V} < V_{th}$$

, which shows it's not in saturation.

$$\text{If } R_S = 12k, V_{GS} = V_G - I_{DS} R_S = 0.6 \text{ V} > V_{th}$$

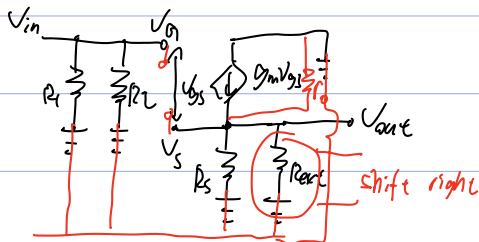
$$V_{DS} = V_{DD} - I_{DS} R_S = 1.8 \text{ V} > V_{GS} - V_{th} = 0.1 \text{ V}$$

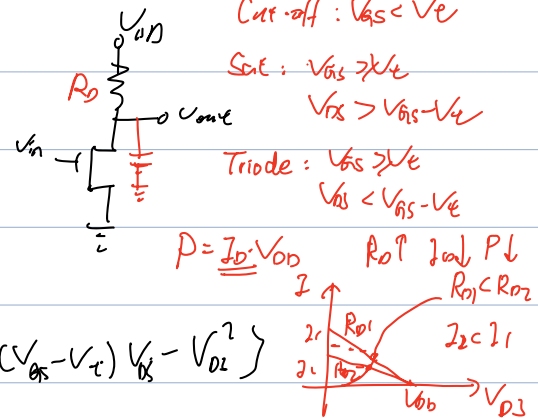
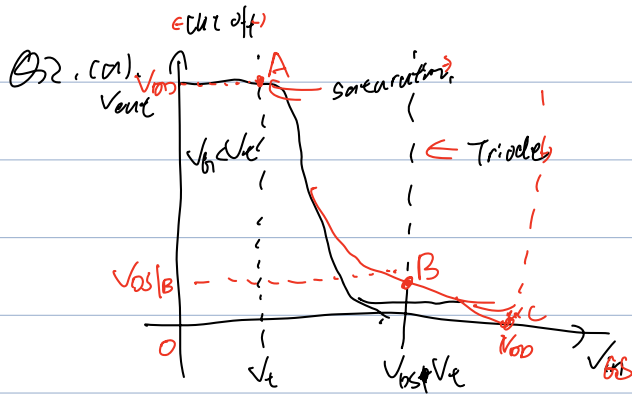
Therefore, $R_S = 12k$.

(c). Assume AC < < < f_T, mid-band frequency, DC short V_{DD} to ground for AC

ideal transistor in saturation

external capacitor short circuit.
internal capacitor open circuit





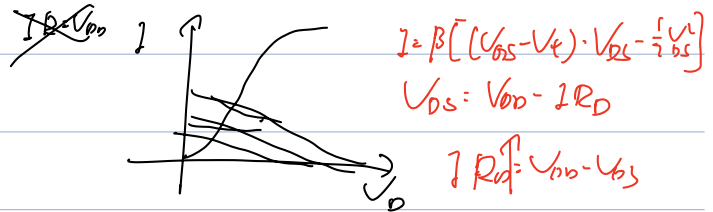
$$I_D = \frac{\beta}{2} \left[2(V_{gs} - V_t)V_{ds} - V_{ds}^2 \right]$$

$$I_D = \frac{\beta}{2} \left[2(V_{gs} - V_t)(V_{DD} - I_{D,sat}R_D) - (V_{DD} - I_{D,sat}R_D)^2 \right]$$

(b)

(c) sat

$$I_D = \frac{V_{DD} - V_D}{R_D}$$



(b) The static power decreases as R_D increases as the load line of the resistor intercepts the MOSFET's current line at a lower value thus $P = I_{D,sat} V_{DD}$ is lower.

(c) The rise time has the biggest impact as it varies with RC whereas the fall time only varies with C .

Q3. (a) $I_{in} = \frac{\mu_0}{2} \frac{W_1}{L_1} (V_{G1} - V_{SS} - V_t)^2$

$$I_{out} = \frac{\mu_0}{2} \frac{W_2}{L_2} (V_{G2} - V_{SS} - V_t)^2$$

As $V_{G1} = V_{G2} \Rightarrow \frac{I_{out}}{I_{in}} = \frac{W_2/L_2}{W_1/L_1} = \frac{W_2 L_1}{W_1 L_2}$

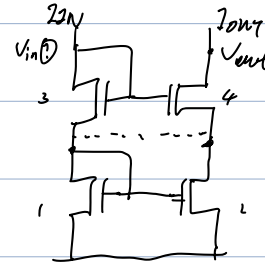
Assuming ideal, $k_1 = k_2$, $V_{t1} = V_{t2}$.

(b) $\frac{I_{out}}{I_{in}} = \frac{W_2/L_2}{W_1/L_1} = \frac{1}{0.5} = 2$

$$\frac{Z_{out}}{b_3} = \frac{W/L_4}{W_3/L_3} = \frac{2/3}{1/3} = 2$$

$$\Rightarrow \frac{Z_{out}}{I_{in}} = 4$$

Cascode
Wilson



$$V_{out} \geq V_{G3} - V_t$$

$$V_{out} = V_{G1} - V_t$$

$$V_{G3} = 2V_{G1} = 2(V_{out} + V_t)$$

$$\Rightarrow V_{out} \geq 2V_{out} + V_t = 2V_{G1} - V_t$$

Q4. (a) Miller's Theorem, Internal capacitors are no longer

open circuit at high frequency.

(b) CMRR = $\frac{A_{diff}}{A_{cm}}$ differential gain divided by common gain
mag of mag of

$$(c). \lambda_{ol} = \frac{-g_m R_G}{1 - \omega^2 L_G + j\omega R_G L_G C_2 \left[\frac{C_1 + C_2}{L_G C_1} - \omega^2 \right]}$$

$$= \frac{-g_m R_G}{1 - \omega^2 L_G + j\omega R_G L_G C_2 \left(\sqrt{\frac{C_1 + C_2}{L_G C_1}} - \omega \right) \left(\sqrt{\frac{C_1 + C_2}{L_G C_1}} + \omega \right)}$$

$$f_0 = \frac{\omega}{2\pi} = \pm \frac{1}{2\pi} \sqrt{\frac{L_G}{C_1 C_2}}$$

Barkhausen Criteria $A\beta \geq 1 + j0$

Use phase condition :

Q5. (a). M_1 is Common Source Amplifier.

M_2 is also Common Source Amplifier,

(b). $V_{G1} = \frac{R_2}{R_1 + R_2} \cdot V_{DD} = \frac{5}{9} \times 5 = \frac{25}{9} \text{ V}$

Assume $AC \ll DC$, in saturation, ideal, then,

$$\begin{cases} I_{DS1} = \frac{\beta}{2} (V_{G1} - V_{S1} - V_t)^2 \\ V_{S1} = I_{DS1} \cdot R_{S1} \end{cases} \Rightarrow I_{DS1} = \frac{\beta}{2} (V_{G1} - I_{DS1} R_{S1} - V_t)^2$$

$\Rightarrow I_{DS1} = 492 \mu\text{A}$ or $660 \mu\text{A}$

if $I_{DS1} = 660 \mu\text{A}$, then $V_{GS1} = V_{G1} - I_{DS1} R_{S1} = 0.138 < V_t$, not saturated.

if $I_{DS1} = 492 \mu\text{A}$, then $V_{GS1} = 0.810 > V_t$

and $V_{DS1} = V_{DD} - I_{DS1} (R_{D1} + R_{S1}) = 0.572 > V_{GS1} - V_t = 0.312$

therefore $I_{DS1} = 492 \mu\text{A}$

(b). $g_{m1} = \sqrt{2\beta I_{DS1}} = \sqrt{2 \times 200 \times 492 \mu} = 3.14 \text{ mS}$

$r_{o1} = \frac{V_A}{I_{DS1}} = \frac{30}{492 \mu} = 60.976 \text{ k}\Omega$

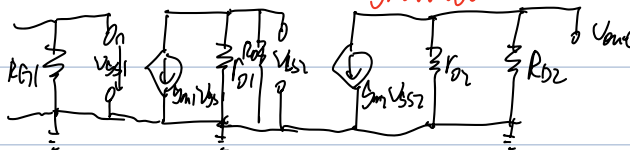
(c). Assume: $AC \ll DC$

mid-band frequency: external capacitors are short circuit

internal capacitors are open circuit

ideal transistors in saturation

Do short for AC to ground



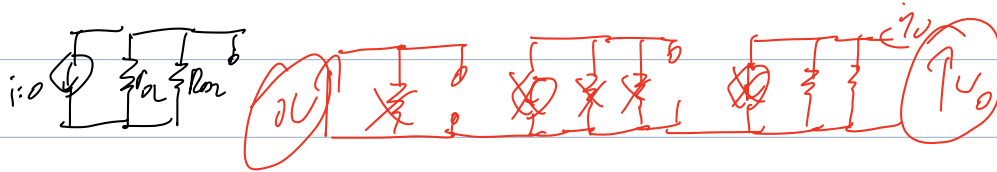
$R_{o1} = R_1 // R_2$

(c). (i) $G_{V1} = -g_{m1} (R_{o1} // R_{D1}) = -4.51$

$G_{V2} = g_{m2} (R_{o2} // R_{D2}) = 3.75$

$$G_M = G_{M1} \cdot G_{M2} = 194.51$$

$$(ii) R_D = r_{D1} \parallel R_D = 4.66 \text{ k}\Omega$$



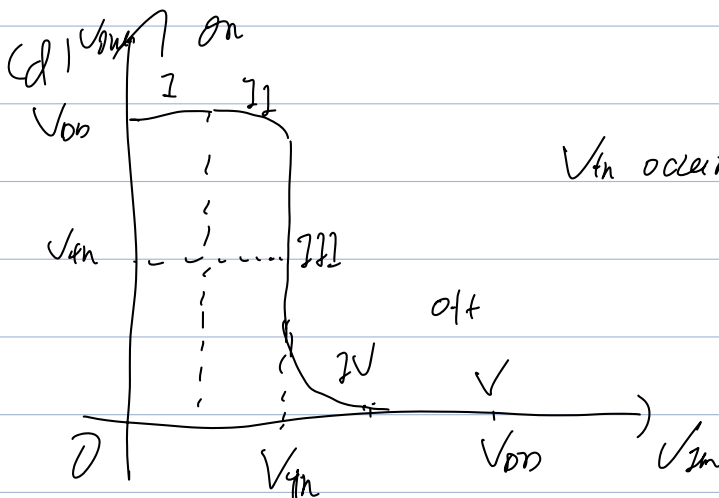
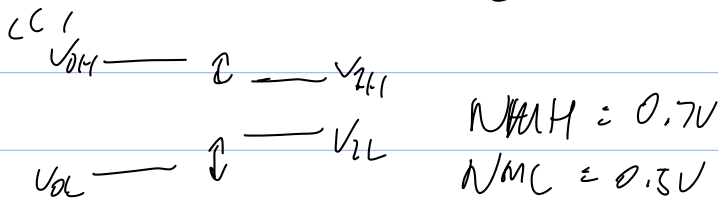
faster switching speed.

Ob. (a). no static current dissipation ideally.

$$(b). P_{static} = V_{DD} \cdot I_{DD} = 3 \times 100 \text{ n} = 300 \text{ nW}$$

$$P_{dynamic} = \underline{C V_{DD}^2 f} = 30 \text{ p} \times 3^2 \times 20 \text{ M} = 5.4 \text{ mW}$$

$$P_{total} = \approx 5.4 \text{ mW}$$



Region	NMOS	PMOS
I	Cut-off	lin
II	Sat	lin
III	Sat	Sat
IV	lin	Sat
V	lin	Cut-off

$$\begin{aligned}
 \text{re). } I_D &= \frac{1}{2} \beta_n (V_{in} - V_{tn})^2 \\
 &= \frac{1}{2} \beta_p (V_{ob} - V_{in} + V_{tp})^2
 \end{aligned}$$

$$V_{th} = V_{tn}$$

$$\Rightarrow \sqrt{\beta_n/\beta_p} (V_{th} - V_{tn}) = (V_{ob} - V_{th} + V_{tp})$$

$$\Rightarrow V_{th} = \frac{V_{ob} + V_{tp} + \sqrt{\beta_n/\beta_p} V_{tn}}{\sqrt{\beta_n/\beta_p} + 1}$$

Q7. ca) (i) ...

(ii),